Textual Data Analysis with Python

via TextBlob

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Overview

- Natural Language Processing (NLP)
- Traditional Language models
- N-grams model
- Bag-of-Words (BoW) model
- TextBlob Library
- How to work with Kaggle
- Hands-on examples/exercise

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Part 1 – Natural Language Processing (NLP)

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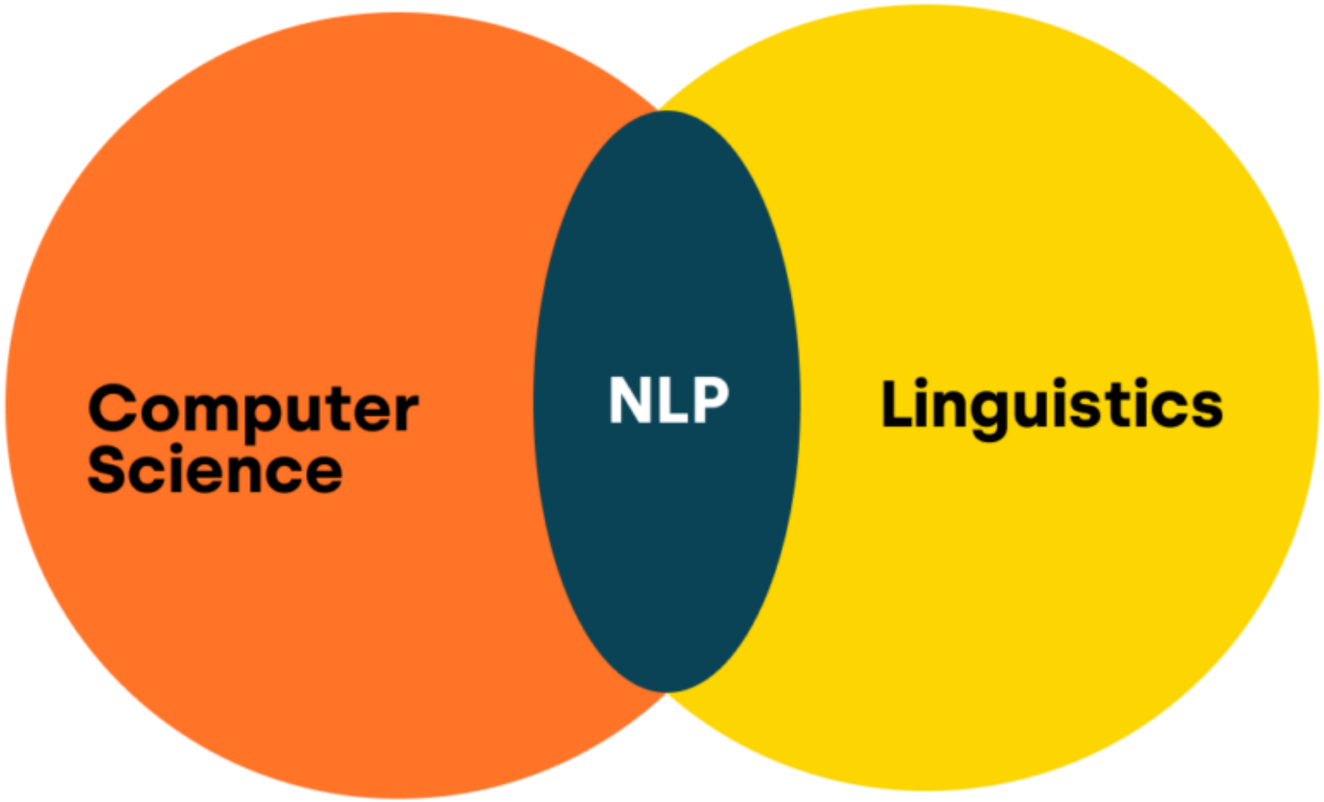
Natural Language Processing (NLP)



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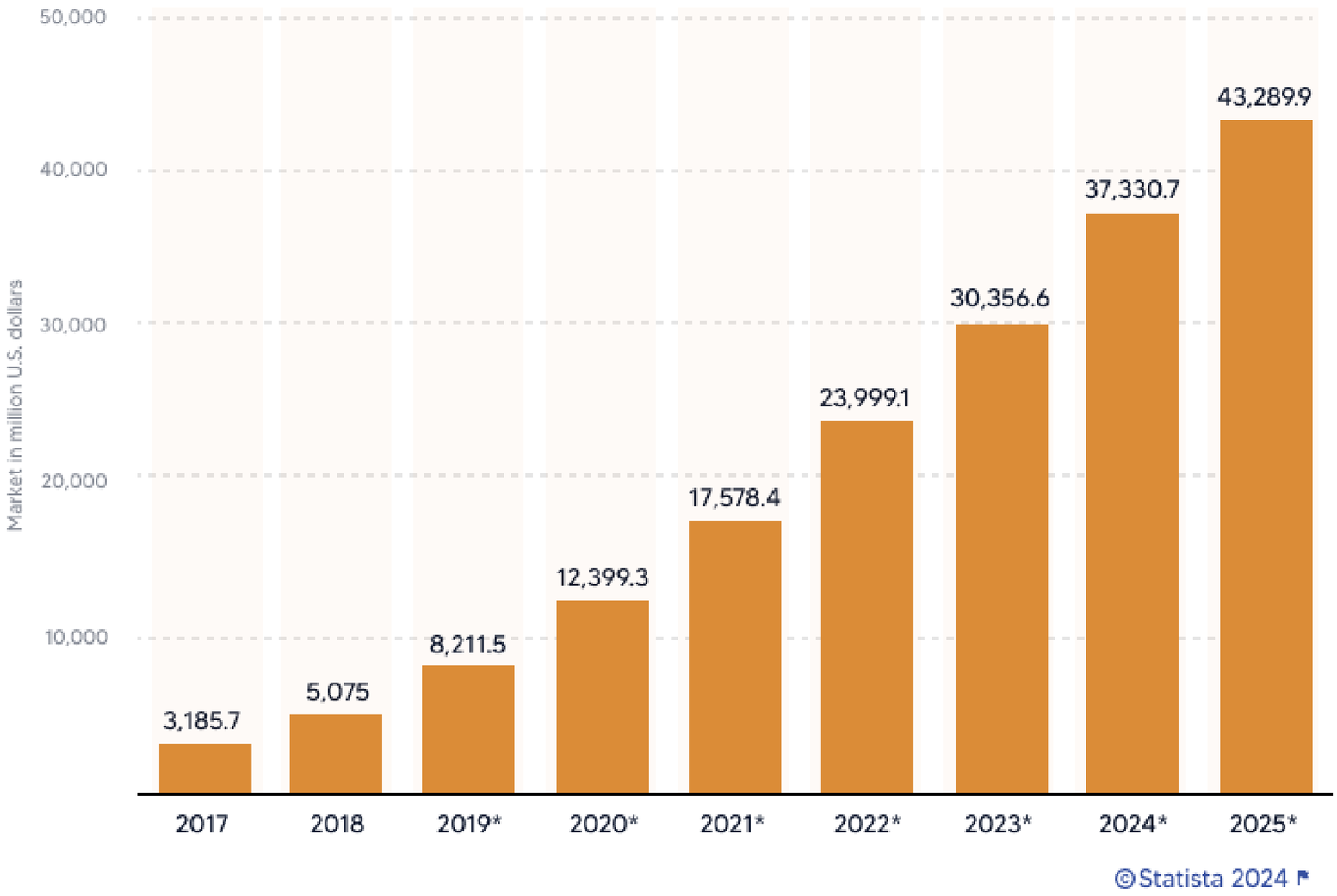
Revenues from the Natural Language Processing (NLP) market worldwide from 2017 to 2025 (in million U.S. dollars)



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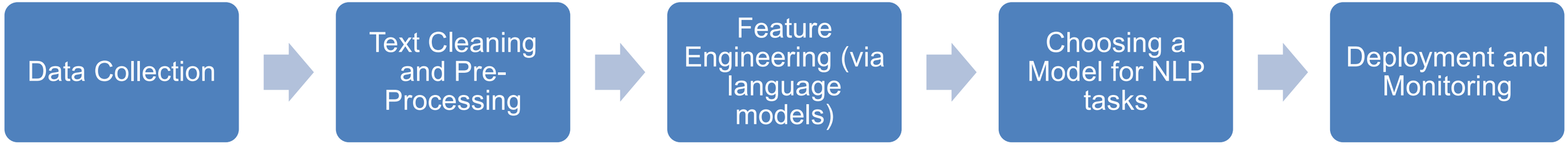
Some applications of NLP:

- Grammer Checkers
- Text Summarization
- Survey Analysis Tools
- Sentiment Analysis Tools
- Language Translators
- Email Filtering and Spam Detection
- Search Autocorrect Tools
- Chatbots
- Hiring Assistants (Resume Checkers)

NLP Project Pipeline:



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NLP text pre-processing techniques:

- Tokenization
- Stop-words removal
- Stemming & Lemmatization
- Part-of-Speech Tagging

Part 2 – Traditional Language Models

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Traditional Language Models

Before LLMs, NLP systems relied on:

- Bag-of-Words (words frequency)
- N-grams (local word order)

Key idea:

- Text was converted into **fixed, context-independent representations**
- Meaning was inferred from **counts, co-occurrence, and handcrafted features**

Traditional Language Models

Bag-of-Words (BoW)

The Bag-of-Words (BoW) model represents text as a collection of words, without considering the order in which they appear. It breaks down a text into individual words, and each word is assigned a count or weight, treating each word as an independent entity.

- **Representation**: Each document is represented as a vector where each entry corresponds to the presence or frequency of a word in that document.
- **Applications**: BoW is often used for document classification, where the task is to classify text based on word occurrences, without caring about the word sequence.

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Traditional Language Models

Bag-of-Words (BoW)

Example: For the sentence "The cat sleeps on the mat", the BoW model would treat it as a collection of individual words: {"The": 2, "cat": 1, "sleeps": 1, "on": 1, "mat": 1}. The word "The" occurs twice, while the other words occur once. The order of words is irrelevant.

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Traditional Language Models

Bag-of-Words (BoW)

We usually visualize it using a matrix called document-term matrix. Example:

Review 1: This movie is very scary and long
Review 2: This movie is not scary and is slow
Review 3: This movie is spooky and good

	1 This	2 movie	3 is	4 very	5 scary	6 and	7 long	8 not	9 slow	10 spooky	11 good	Length of the review(in words)
Review 1	1	1	1	1	1	1	1	0	0	0	0	7
Review 2	1	1	2	0	0	1	1	0	1	0	0	8
Review 3	1	1	1	0	0	0	1	0	0	1	1	6

Vector-of-Review1: [1 1 1 1 1 1 1 0 0 0 0]

Vector-of-Review2: [1 1 2 0 0 1 1 0 1 0 0]

Vector-of-Review3: [1 1 1 0 0 0 1 0 0 1 1]

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Traditional Language Models

Bag-of-Words (BoW)

Limitations:

Loss of context: BoW models ignore word order and syntax, which can lead to poor performance in tasks where word sequence matters.

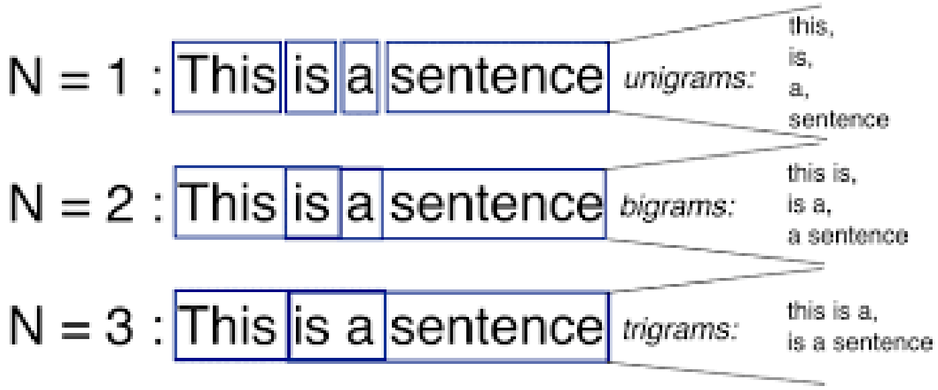
Large vocabulary size: As the vocabulary grows, the dimensionality of the model increases, making it less efficient.

Traditional Language Models

N-Grams

An n-gram model is one of the simplest traditional language models. It predicts the next word in a sequence based on the preceding n-1 words. The idea is that the probability of a word depends on the context of the last few words, with n representing the number of words considered.

- **Unigram**: Only the current word is considered (no context).
- **Bigram**: The model considers one previous word.
- **Trigram**: The model considers two previous words, and so on.



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Traditional Language Models

N-Grams

Example: In a bigram model, given the sentence "The cat is", the model would predict the next word by looking at the probability of each word following "is" based on its training data.

The sentence's probability is calculated by multiplying the conditional probabilities of each word given the previous one:

$$P(\text{The cat is cute}) = P(\text{The}) \times P(\text{cat}|\text{The}) \times P(\text{is}|\text{cat}) \times P(\text{cute}|\text{is})$$

It means to get the probability of the whole sequence, you multiply the probability of the first word, by the probability of the second word given the first, by the probability of the third word given the first two, and so on.

Here for example P (cat|The) is a conditional probability of the word "cat" appearing given that the word "The" has already appeared.

The vertical bar "|" means given.

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Traditional Language Models

N-Grams

Limitations:

Limited context: n-gram models only capture local dependencies and cannot handle long-range dependencies between words.

Data sparsity: If certain n-grams don’t appear in the training data, the model can’t make predictions for them, requiring smoothing techniques to deal with unseen word combinations.

Traditional Language Models

Overall Challenges and limitations

- Limited context understanding
- Data sparsity issues
- Vocabulary limitations
- Manual feature engineering
- Scalability constraints
- Inability to handle long-range dependencies
- Task-specific nature
- Poor adaptability
- Sequential data processing

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Part 3 – TextBlob

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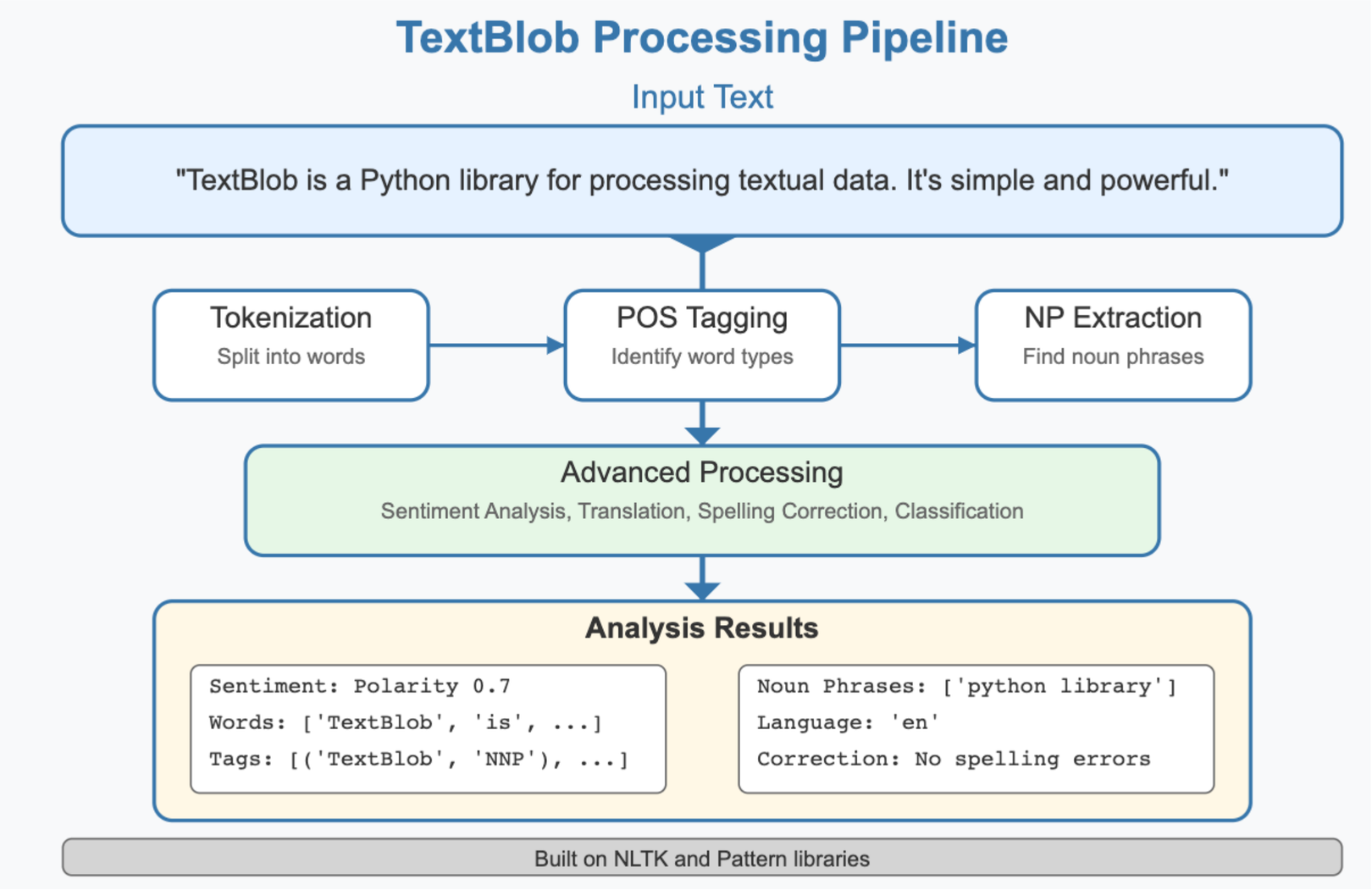
TextBlob



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Part 4 – Kaggle

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Kaggle Notebooks

Kaggle Notebooks are essentially Jupyter Notebooks hosted on the Kaggle platform (on the cloud).

- They share the same interactive environment as Jupyter Notebooks, allowing you to combine code, visualizations, and text within a single document.
- You don't need to set up your own environment. Kaggle provides the computational resources (GPUs, TPUs) and a convenient interface for running your code in the cloud.
- Kaggle has a large and active community where you can share and collaborate.
- Kaggle hosts numerous competitions where you can test your skills and collaborate with others.

Create an account if you don't have one already, it is free to create a Kaggle account.

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Part 5 – The hands-on examples/exercises

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